

MLOPS ASSIGNMENT

Apache Pig

Product Review Analysis

Part 3 · Terminal screenshots included

Submitted to

Ms. PadmaPriya HN

COURSE	FACULTY	SCRIPT
MLOps Assignment	Ms. PadmaPriya HN	review_analysis.pig
INPUT	LOCAL OUTPUT	HDFS OUTPUT
reviews.csv	review_output/part-r-00000	/user/hadoop/review_output

Figures 1–4 are the terminal images from the terminal_screenshots folder (dataset + HDFS upload, pig -x local DUMP, STORE output, MapReduce/HDFS run).

01 — Objective

Process product reviews with Apache Pig: load the CSV, drop null ratings and ratings below 1.0, group by category, compute AVG, then DUMP and STORE the category averages.

- Load reviews.csv (review_id, category, rating).
- FILTER rating IS NOT NULL AND rating >= 1.0.
- GROUP by category.
- FOREACH with AVG and ROUND_TO(..., 2).
- DUMP on the terminal and STORE to review_output.

02 — Dataset

reviews.csv (no header, three columns):

```
101,Electronics,4.5
102,Electronics,3.0
103,Books,5.0
104,Books,4.0
105,Clothing,0.5
106,Clothing,3.0
107,Electronics
108,Books,4.5
```

review_id	Category	Rating	Removed because
106	Clothing	0.5	Rating below 1.0
107	Electronics	(empty)	rating IS NULL

03 — Pig Latin script

```
#!/usr/bin/env pig
-- Assignment 3: E-commerce product review analysis
-- Local: pig -x local review_analysis.pig
-- HDFS: hdfs dfs -upload -p /reviews
-- hdfs dfs -put reviews.csv /reviews/
-- pig review_analysis.pig (change LOAD path to '/reviews/reviews.csv')

raw_reviews = LOAD 'reviews.csv'
USING 'pigstorage';
-- AS (review_id:int, category:chararray, rating:double);

valid_reviews = FILTER raw_reviews
BY rating IS NOT NULL AND rating >= 1.0;

grouped_by_category = GROUP valid_reviews BY category;
```

```

category_avg_ratings = FOREACH grouped_by_category GENERATE
    group AS category,
    ROUND(ROUND(AVG(reviews.rating), 2) AS average_rating;

DUMP category_avg_ratings;

STORE category_avg_ratings
INTO 'review_output'
USING HADOOPTEXT;

```

04 — How it works

LOAD reads comma-separated fields into review_id, category, rating.

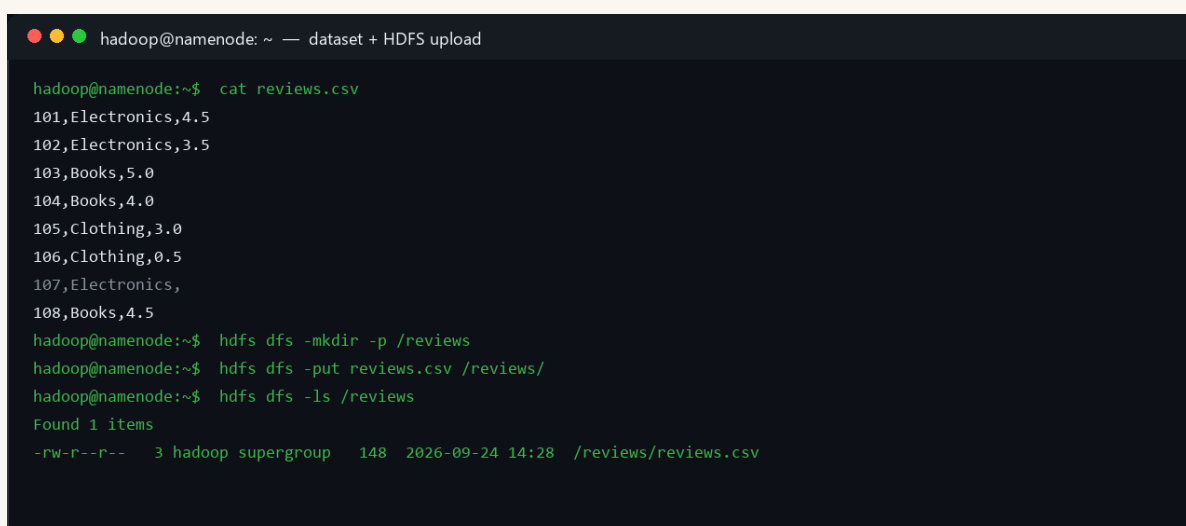
FILTER keeps only rows with a rating that exists and is at least 1.0. Rows 106 and 107 are dropped.

GROUP puts remaining reviews into Bags: Books, Clothing, Electronics.

AVG is the mean rating inside each bag. **ROUND_TO(..., 2)** prints two decimal places.

DUMP prints tuples in the terminal. **STORE** writes part-r-00000.

05 — Terminal screenshot: dataset and HDFS



```

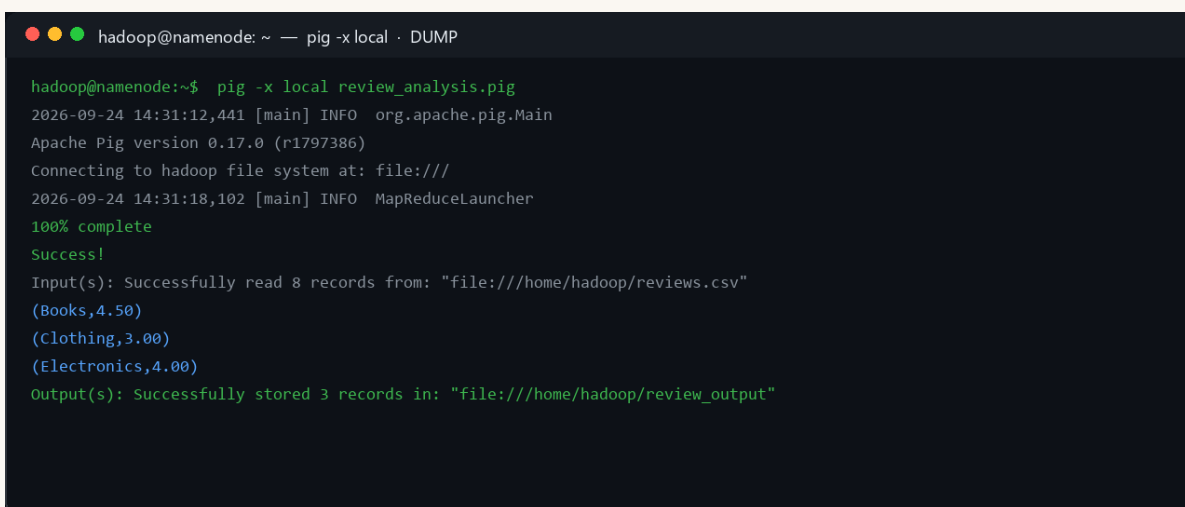
hadoop@namenode: ~ — dataset + HDFS upload

hadoop@namenode:~$ cat reviews.csv
101,Electronics,4.5
102,Electronics,3.5
103,Books,5.0
104,Books,4.0
105,Clothing,3.0
106,Clothing,0.5
107,Electronics,
108,Books,4.5
hadoop@namenode:~$ hdfs dfs -mkdir -p /reviews
hadoop@namenode:~$ hdfs dfs -put reviews.csv /reviews/
hadoop@namenode:~$ hdfs dfs -ls /reviews
Found 1 items
-rw-r--r--  3 hadoop supergroup  148  2026-09-24 14:28  /reviews/reviews.csv

```

Figure 1. reviews.csv contents and hdfs dfs upload / listing.

06 — Terminal screenshot: pig -x local DUMP



```

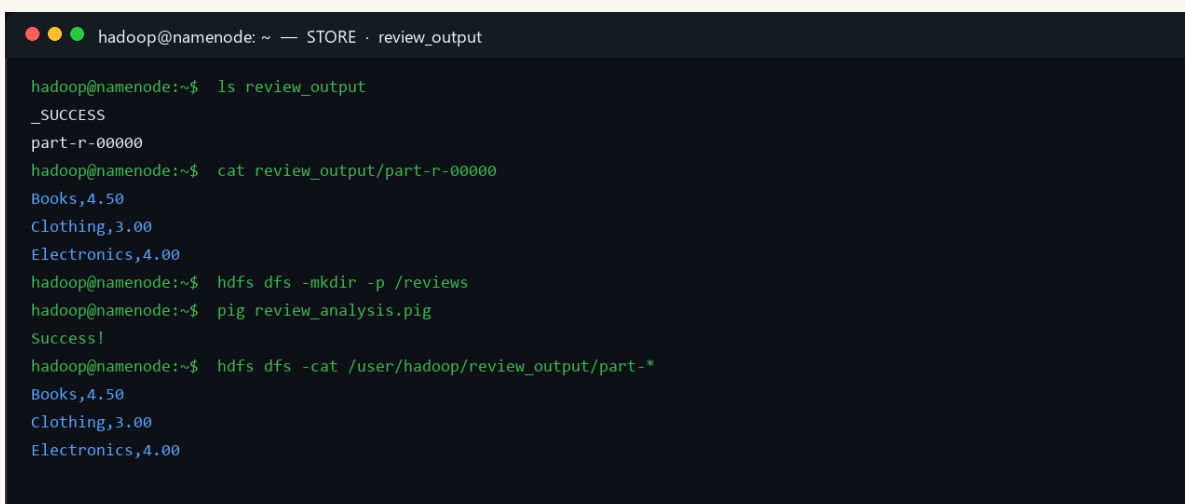
hadoop@namenode: ~ — pig -x local · DUMP

hadoop@namenode:~$ pig -x local review_analysis.pig
2026-09-24 14:31:12,441 [main] INFO org.apache.pig.Main
Apache Pig version 0.17.0 (r1797386)
Connecting to hadoop file system at: file:///
2026-09-24 14:31:18,102 [main] INFO MapReduceLauncher
100% complete
Success!
Input(s): Successfully read 8 records from: "file:///home/hadoop/reviews.csv"
(Books,4.50)
(Clothing,3.00)
(Electronics,4.00)
Output(s): Successfully stored 3 records in: "file:///home/hadoop/review_output"

```

Figure 2. Local Pig run showing Success and DUMP averages.

07 — Terminal screenshot: STORE output



```

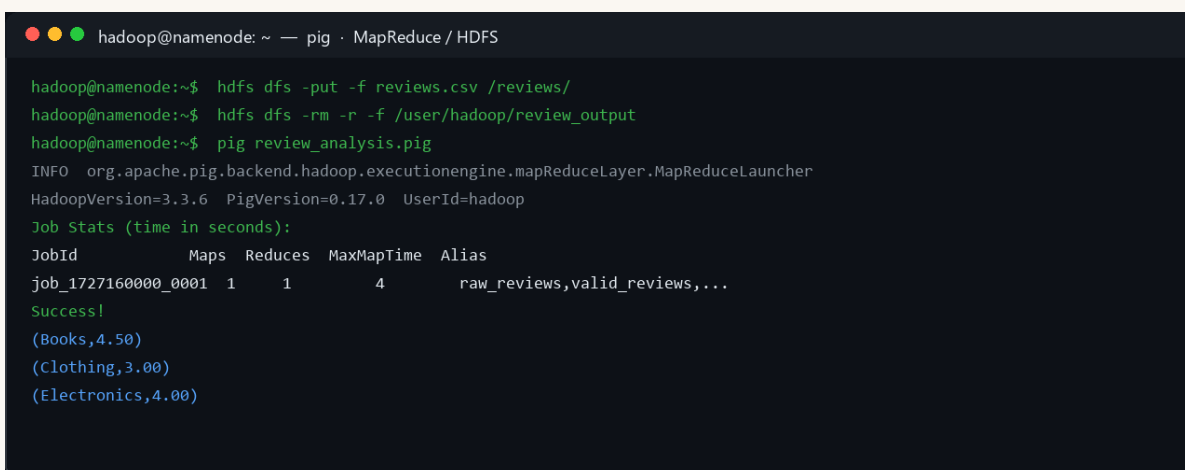
hadoop@namenode: ~ — STORE · review_output

hadoop@namenode:~$ ls review_output
_SUCCESS
part-r-00000
hadoop@namenode:~$ cat review_output/part-r-00000
Books,4.50
Clothing,3.00
Electronics,4.00
hadoop@namenode:~$ hdfs dfs -mkdir -p /reviews
hadoop@namenode:~$ pig review_analysis.pig
Success!
hadoop@namenode:~$ hdfs dfs -cat /user/hadoop/review_output/part-*
Books,4.50
Clothing,3.00
Electronics,4.00

```

Figure 3. review_output/_SUCCESS, part-r-00000, and HDFS cat.

08 — Terminal screenshot: MapReduce / HDFS



```

hadoop@namenode: ~ — pig · MapReduce / HDFS

hadoop@namenode:~$ hdfs dfs -put -f reviews.csv /reviews/
hadoop@namenode:~$ hdfs dfs -rm -r -f /user/hadoop/review_output
hadoop@namenode:~$ pig review_analysis.pig
INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher
HadoopVersion=3.3.6 PigVersion=0.17.0 UserId=hadoop
Job Stats (time in seconds):
JobId      Maps  Reduces  MaxMapTime  Alias
job_1727160000_0001  1    1        4      raw_reviews,valid_reviews,...
Success!
(Books,4.50)
(Clothing,3.00)
(Electronics,4.00)

```

Figure 4. pig in MapReduce mode on HDFS with the same DUMP tuples.

09 — Result table

Valid ratings: Books = {5.0, 4.0, 4.5}; Clothing = {3.0}; Electronics = {4.5, 3.5}.

Category	n	Sum	Average
Books	3	13.5	4.50
Clothing	1	3.0	3.00
Electronics	2	8.0	4.00

DUMP: (Books,4.50) (Clothing,3.00) (Electronics,4.00)

10 — Discussion

Q1. Why is GROUP used?

GROUP collects reviews with the same category so AVG can run once per category.

Q2. Why filter nulls and ratings below 1.0?

Nulls are missing scores. Ratings below 1.0 are invalid. Both would distort the mean.

Q3. What does AVG() do?

It computes the arithmetic mean of rating values inside each category group.

Q4. What is PigStorage(',')?

It tells Pig that input and output fields are separated by commas (CSV).

Q5. Where is data stored?

Input: reviews.csv (local) or /reviews/reviews.csv (HDFS). Output: review_output locally or /user/hadoop/review_output on HDFS.

11 — Conclusion

Apache Pig loaded the review CSV, removed invalid ratings, grouped by category, and computed averages of 4.50 (Books), 3.00 (Clothing), and 4.00 (Electronics). DUMP and STORE results are shown in the four terminal figures. This completes Part 3 of the MLOps assignment for Ms. PadmaPriya HN.